

Automotive Sensors

LiDAR

Automotive Intelligence Lab.



HANYANG UNIVERSITY

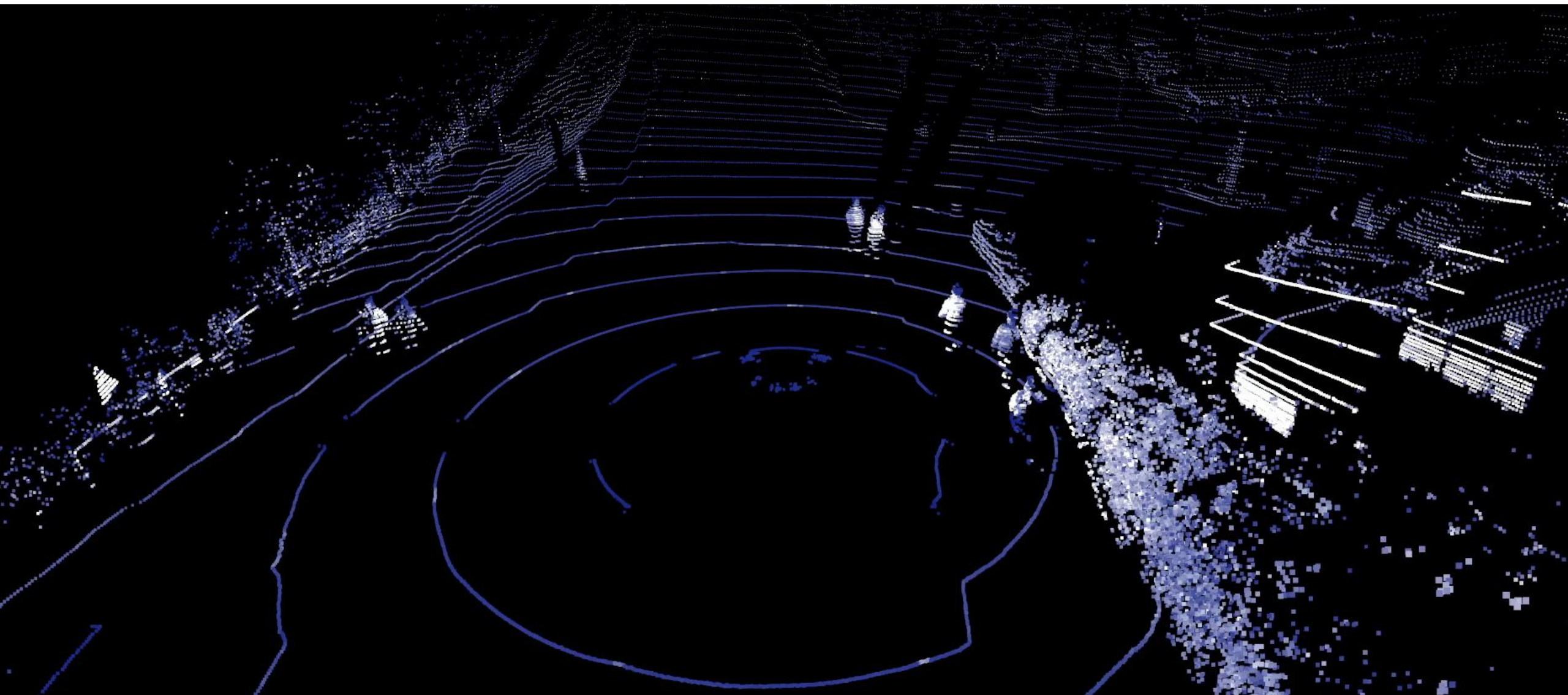


Contents

- Principle of LiDAR
- LiDAR based perception system

Principle of LiDAR

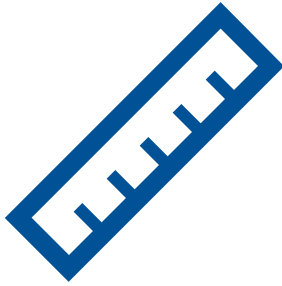
LiDAR (Light Detection And Ranging)



LiDARs on Autonomous Cars



Classification Criteria of LiDAR sensors



Distance measurement



3D Beam-scanning
technology



Wavelength of light

1. Distance measurement

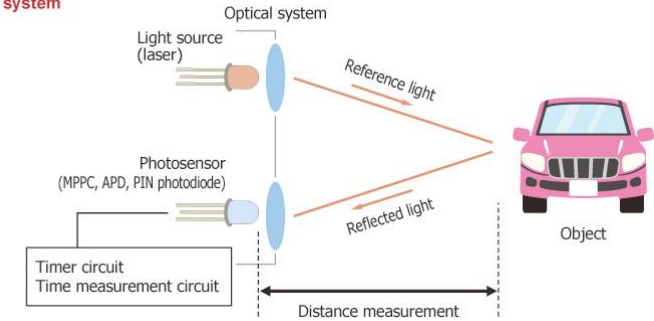


LiDAR

Direct
Detection

ToF

TOF system

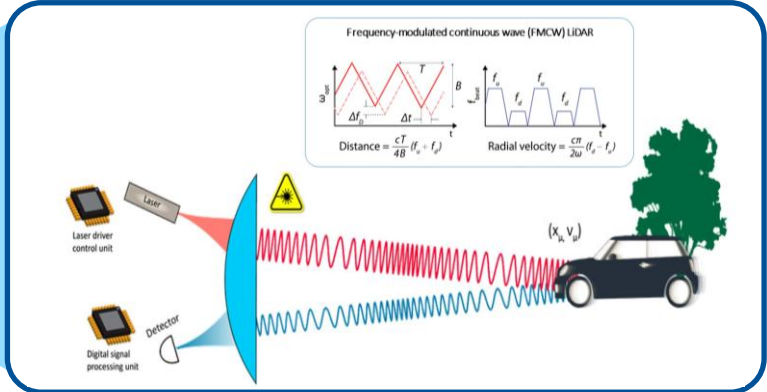


Indirect
Detection

FMCW

AMCW

Etc.



Trigonometry

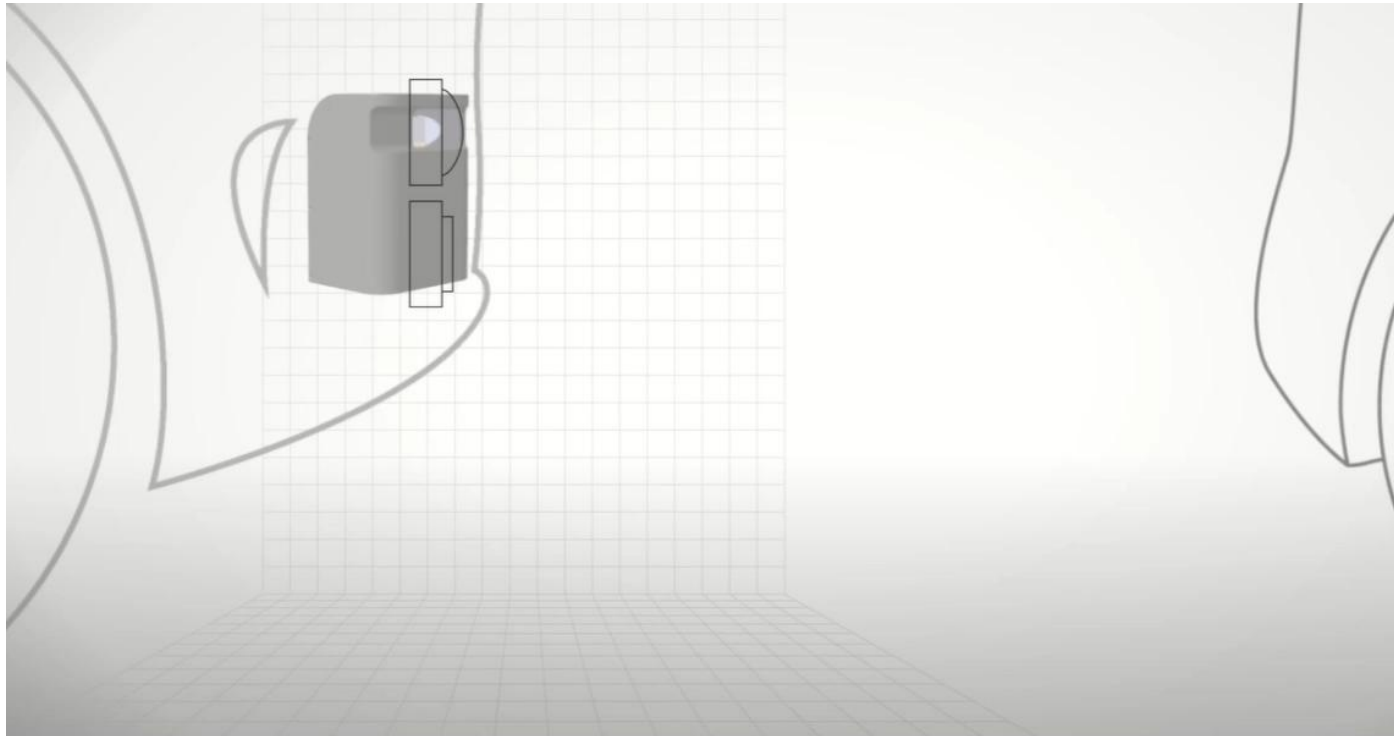
Phase

- ToF: Time of Flight
- FMCW: Frequency Modulation Continuous Wave
- AMCW: Amplitude Modulation Continuous Wave

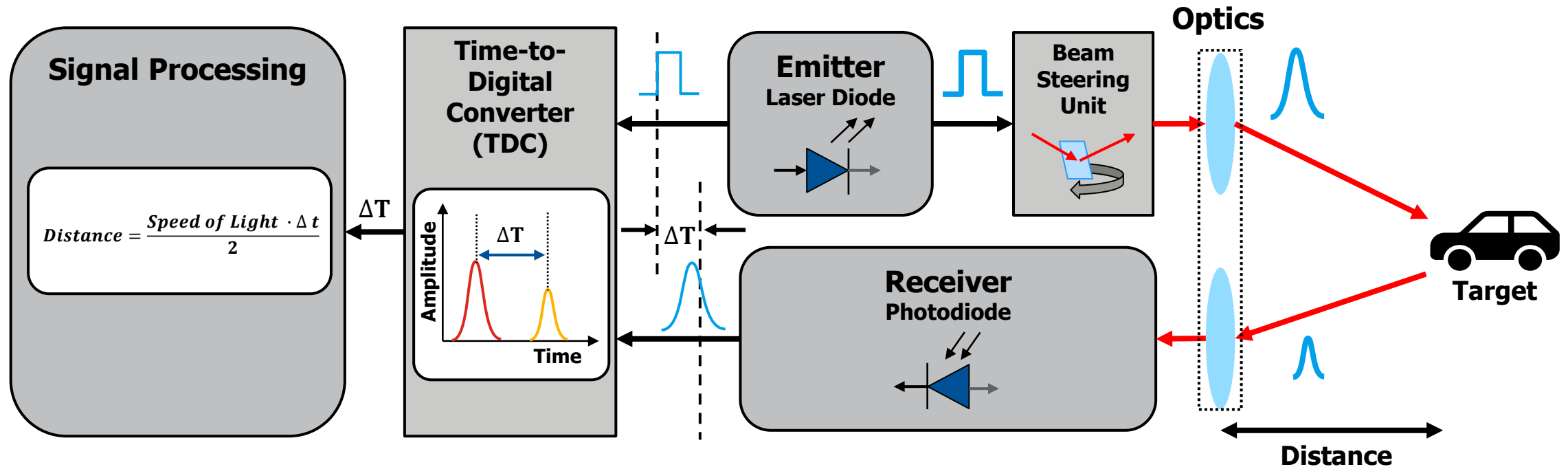
ToF (Time of Flight)

■ Time of Flight (ToF)

- ▶ Emitting a laser pulse on a surface
- ▶ Catching the echo pulse to the LiDAR pulse source with sensors
- ▶ Measuring the time laser travelled using TDC (time to digital converter) threshold



LIDAR structure



Reflectivity and Intensity

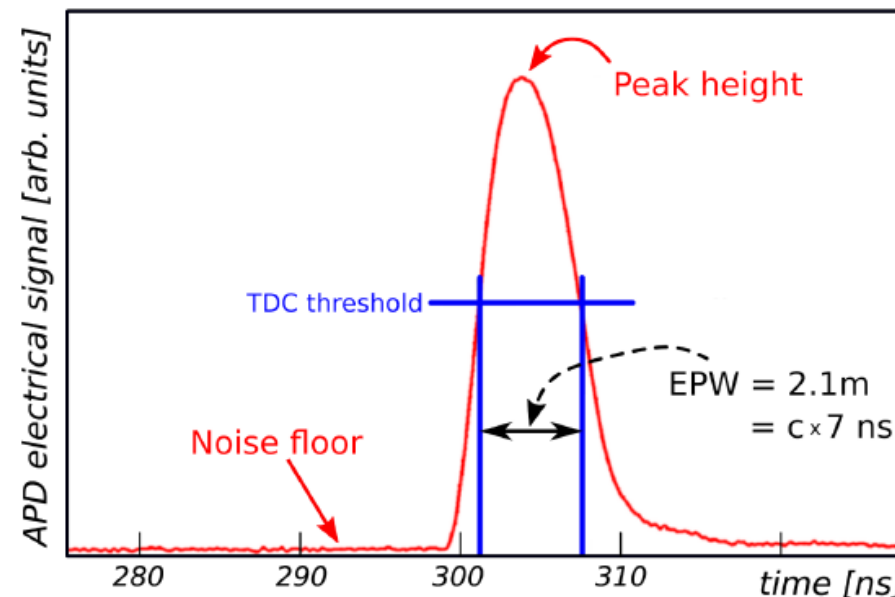
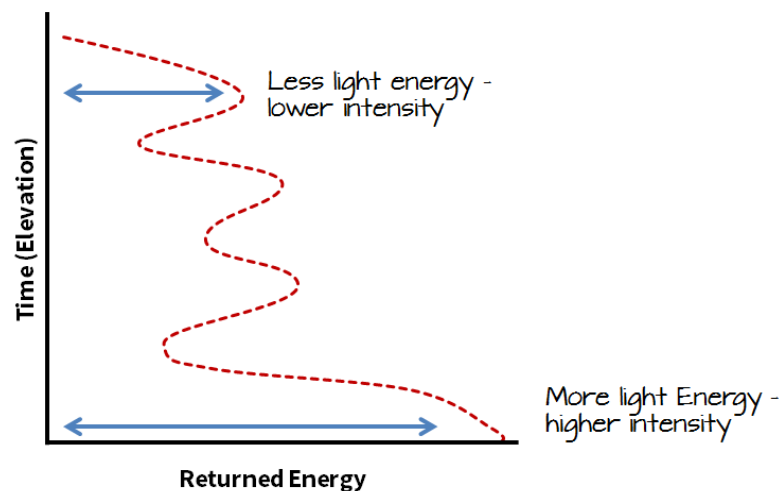
■ Reflectivity

- ▶ The strength of the reflectivity varies with the composition of the surface object reflecting the return.



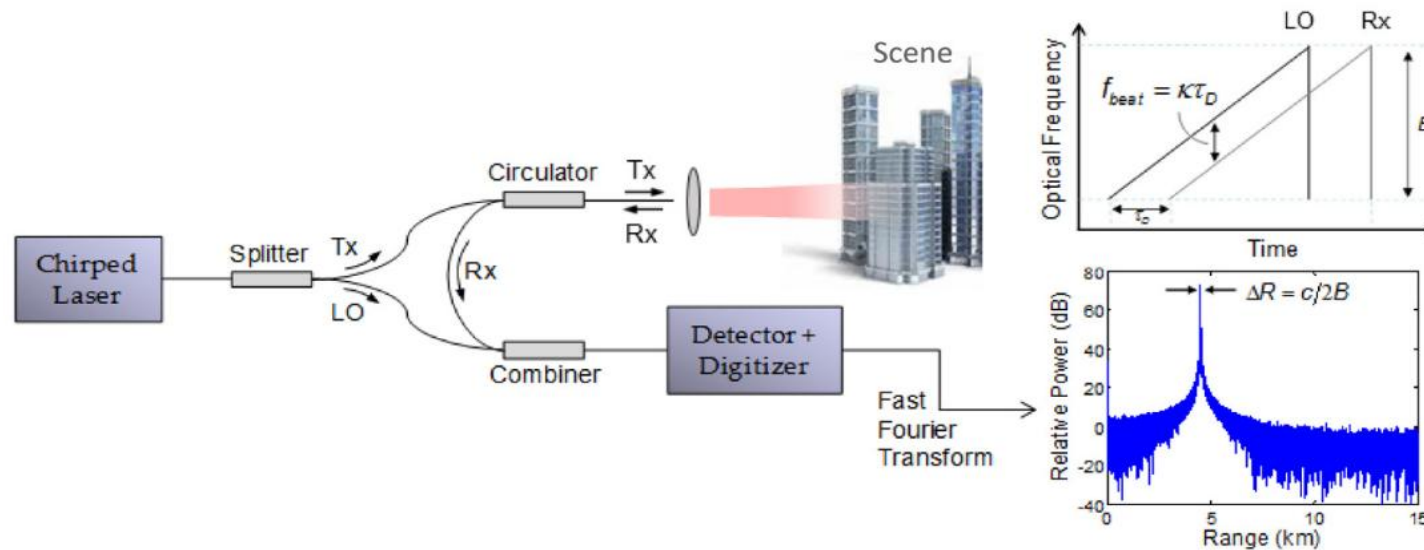
■ Intensity

- ▶ Measurement collected for every point of the return strength of the laser pulse.
- ▶ Based in part on the reflectivity of the object hit by the laser pulse.



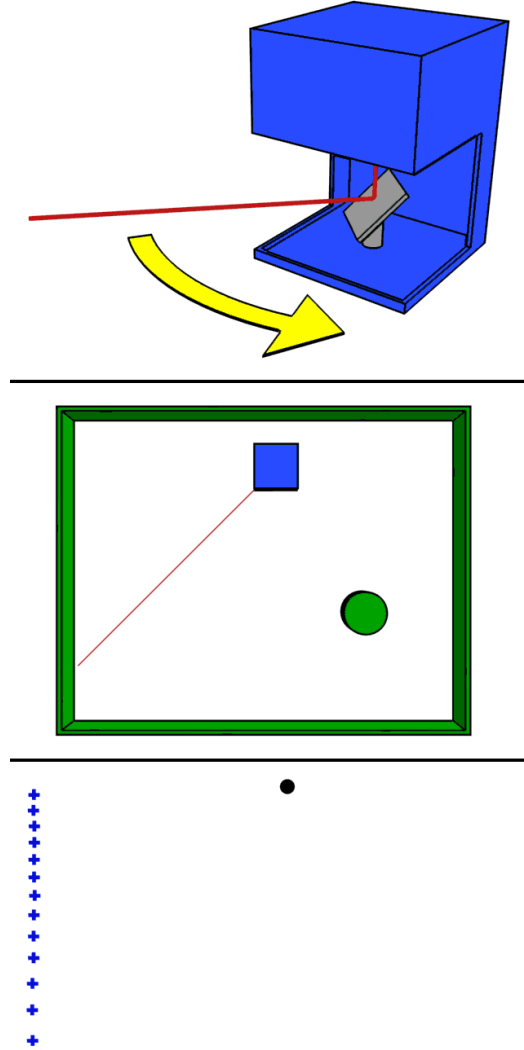
FMCW (Frequency-Modulated Continuous Wave)

- FMCW lidar sends out a **constant stream of light** (“continuous-wave”) and **changes the frequency of that light at regular intervals** (“frequency-modulated”).
 - ▶ This allows us to both determine the location of objects and precisely measure their velocity using the Doppler effect.
- Due to coherent detection, FMCW technology can achieve
 - ▶ **high resolution** without requiring fast electronics or high optical power,
 - ▶ **immune to direct sun light** and **interference from other LiDAR transmitters**.

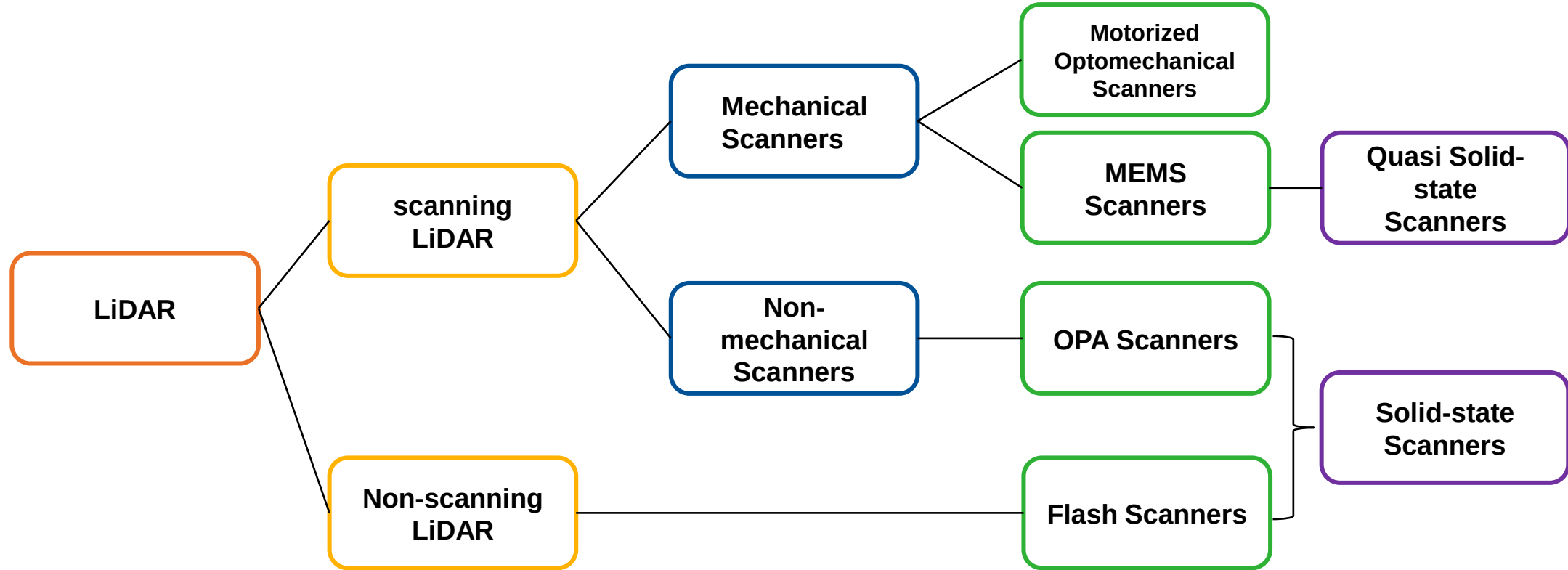


FMCW (Frequency-Modulated Continuous Wave)

2. 3D Beam-scanning technology



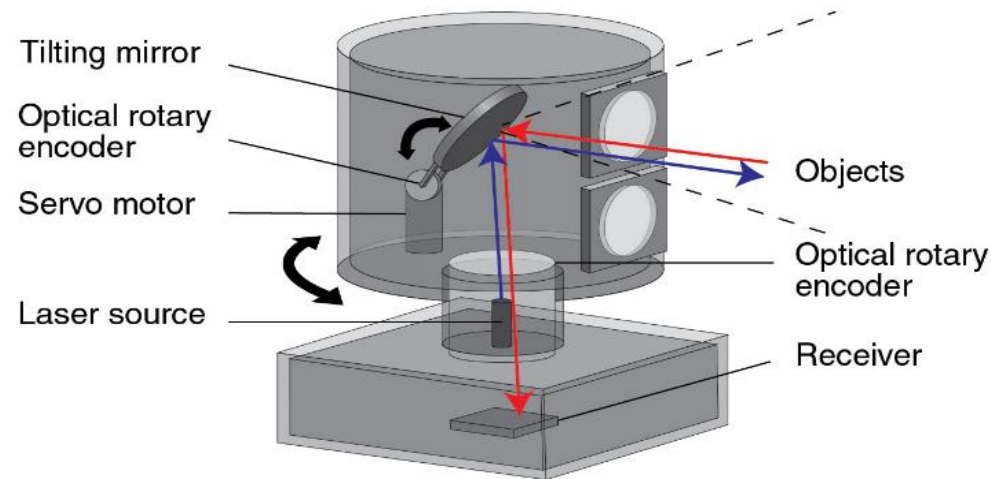
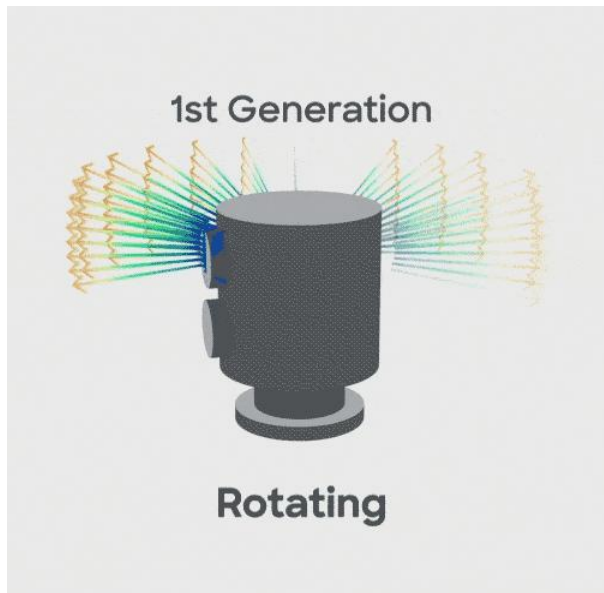
2. 3D Beam-scanning technology



Mechanical LiDAR

■ Mechanical(Motorized) LiDARs

- ▶ Motorized optomechanical scanners are the **most common type** of LiDAR Scanners.
- ▶ Motorized LiDAR rotates **multiple channels of transmitters and receivers** by a motor.
- ▶ These kind of LiDARs are **not power-efficient** and are **vulnerable** to mechanical shock and wear.
- ▶ Their vertical resolution is **fixed** and dependent on the number of transmitter and receiver channels, so a **high vertical resolution is always at the price of high cost**.



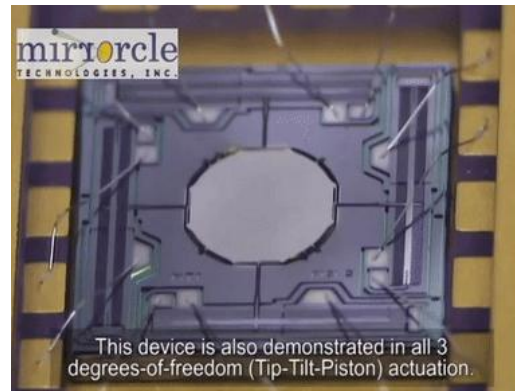
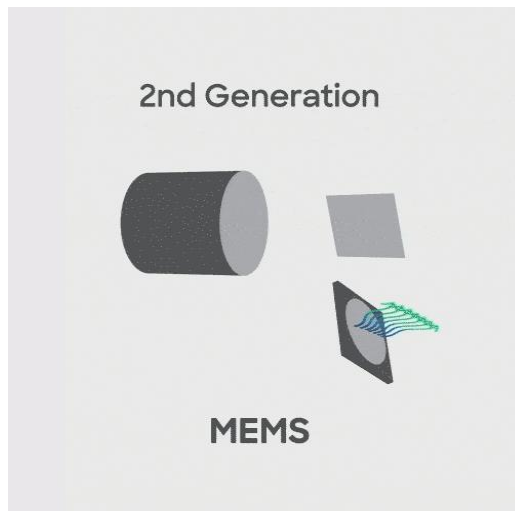
Mechanical LiDAR



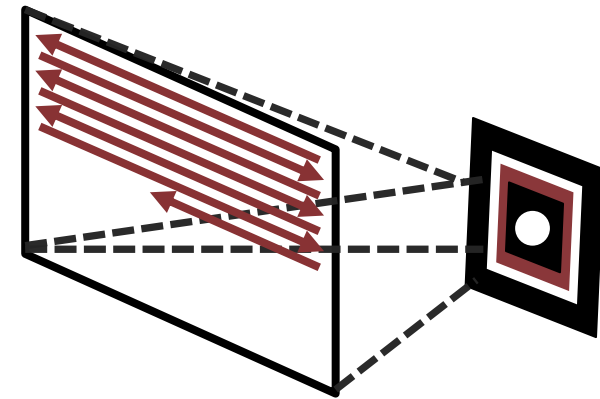
LiDAR Beam captured by IR Camera

MEMS (Micro-Electro-Mechanical Systems) LiDAR

- Superior in terms of **size**, **scanning speed**, and **cost**.
- **Tilting tiny mirrors** to precisely steer the laser beam.

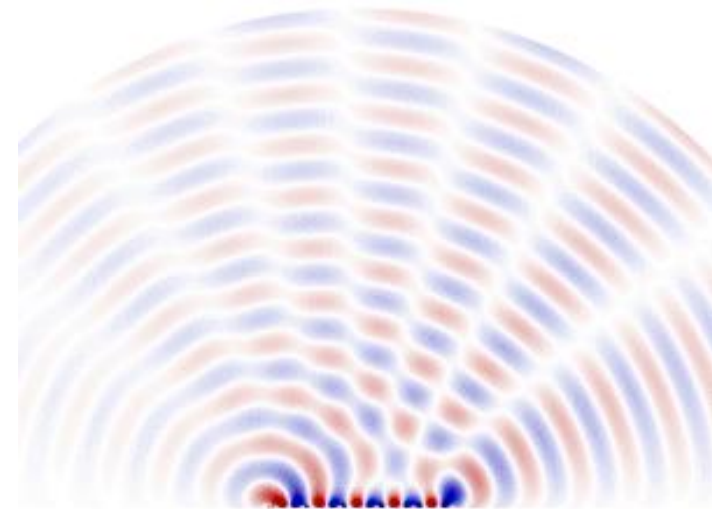
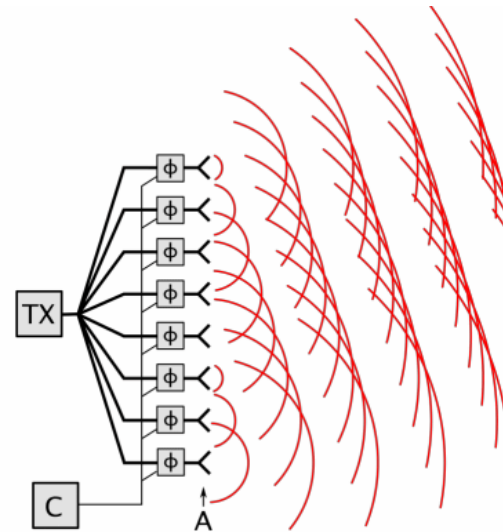
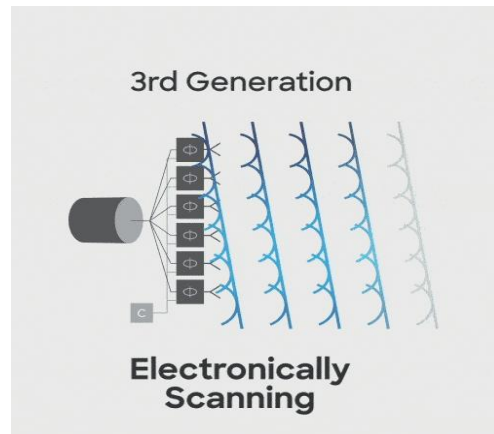


MEMS 3D Scan



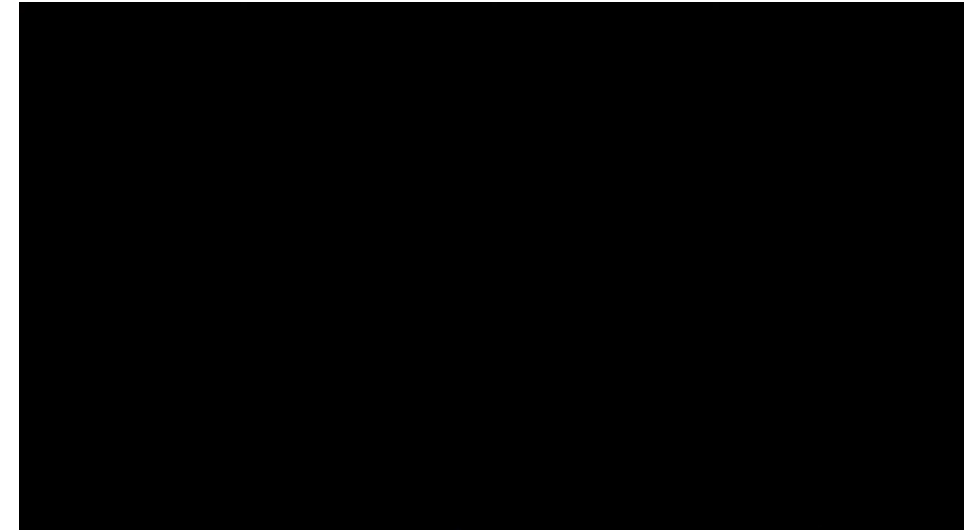
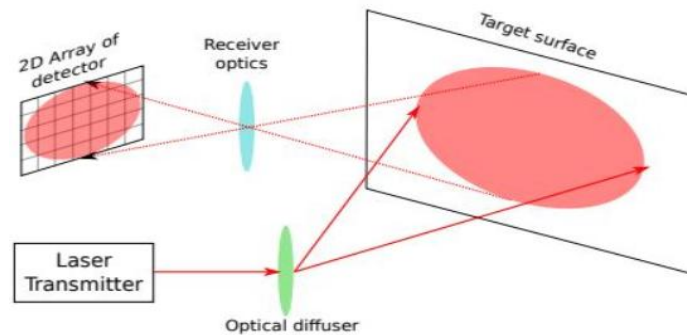
OPA LiDAR

- Superior in terms of **stabilization**, **size**, **cost** and **reliability**.
- Steering the laser beam by **adjusting the phase** of the light emitted, **without moving parts**.



Flash LiDAR

- Superior in terms of **vibrations resistant**, **scanning speed**, **size**, and **price**.
- Like flash camera, **capturing the entire scene** quickly in **single flash**.
- **High laser power** is needed to illuminate the entire scene and detect at a distance.

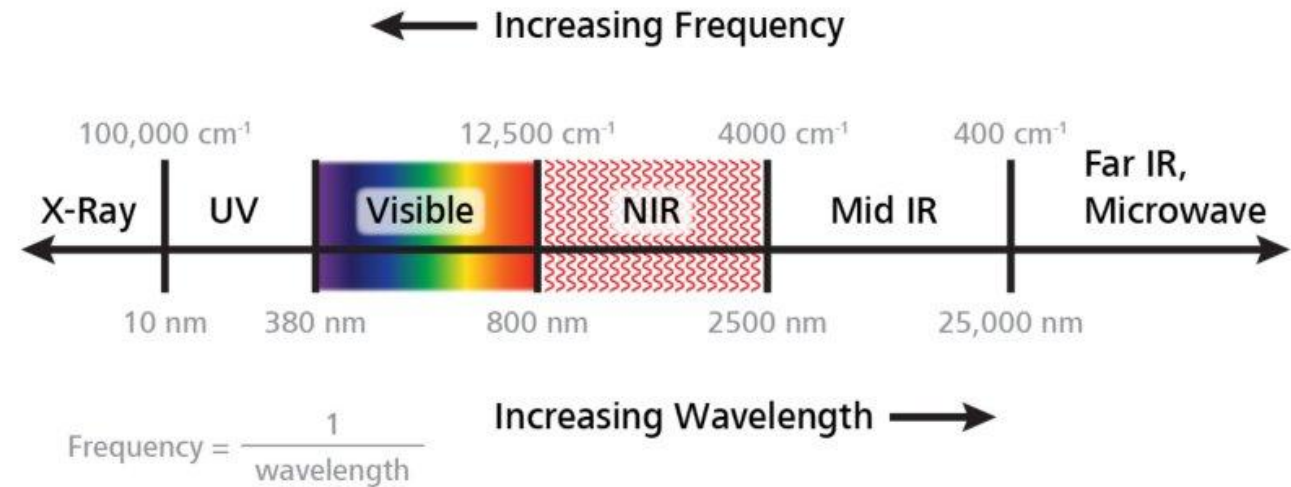


3. Wavelength of Light

■ The types of wavelengths and laser types used in laser beams are as follows:

▶ Spectrum range: Near-InfraRed(NIR)

- 850nm
 - Ouster
- 905nm
 - Velodyne
- 1550nm
 - Luminar



850nm LiDAR Ouster



905nm LiDAR Velodyne

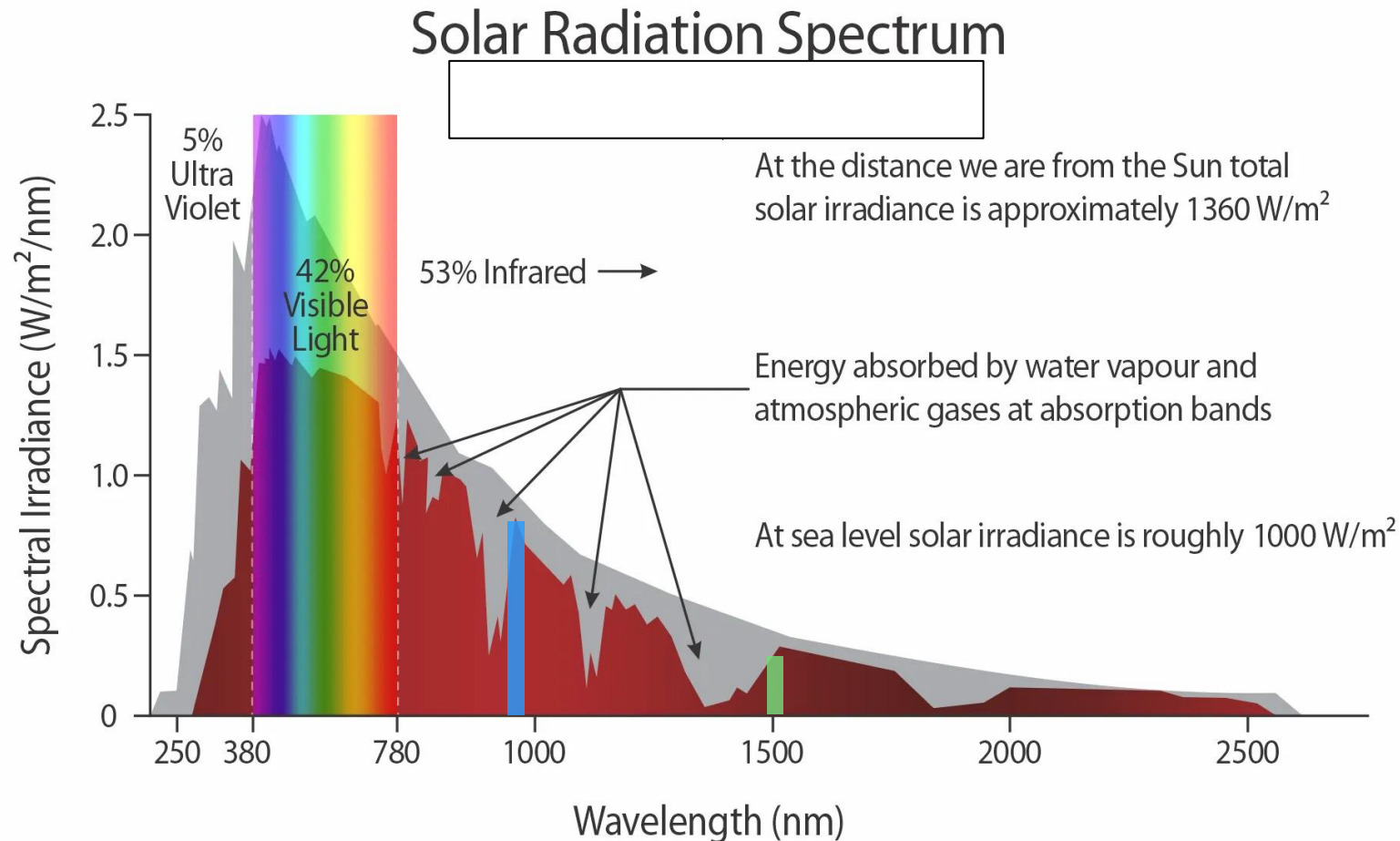


1550nm LiDAR Luminar

Wavelength

■ Comparison of frequently used wavelengths of 905 nm with higher power of 1550 nm.

- ▶ *Background noise*
- ▶ 905nm has 3 times more background noise than 1550nm



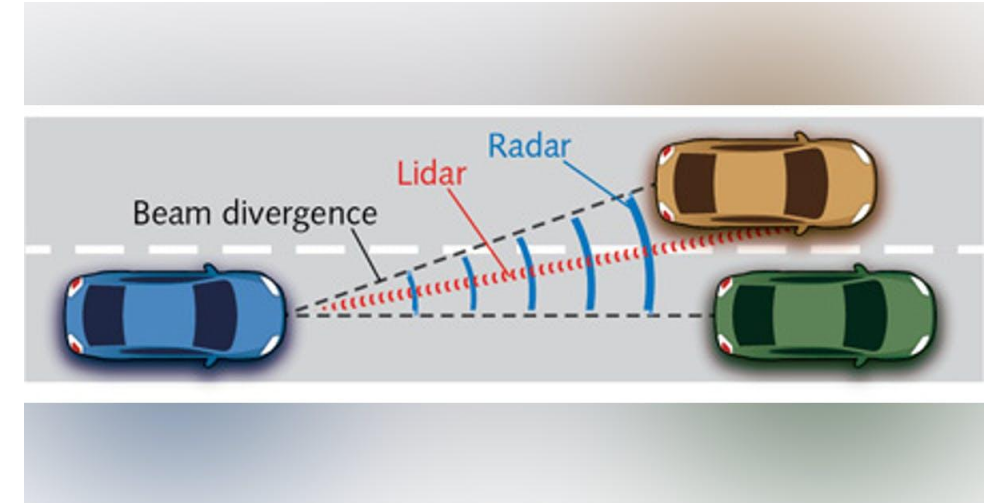
Beam Divergence

■ Beam divergence:

- ▶
- ▶
- ▶ The longer the wavelength, the more beam spreads

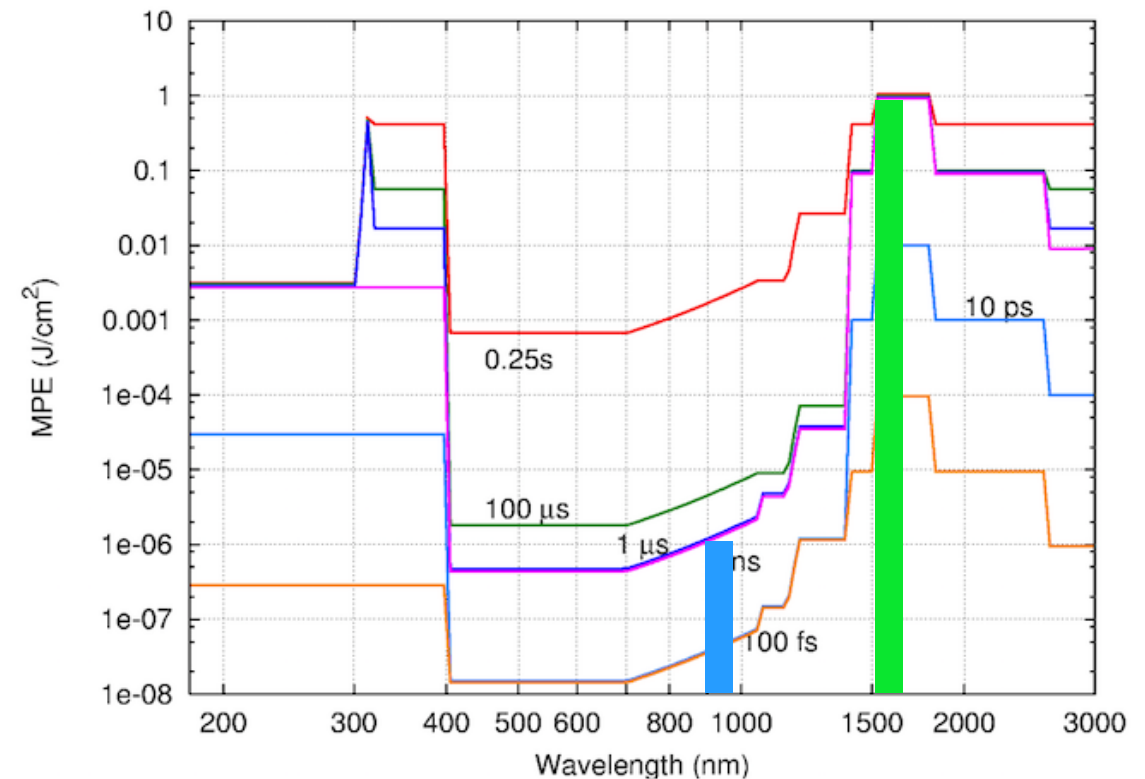
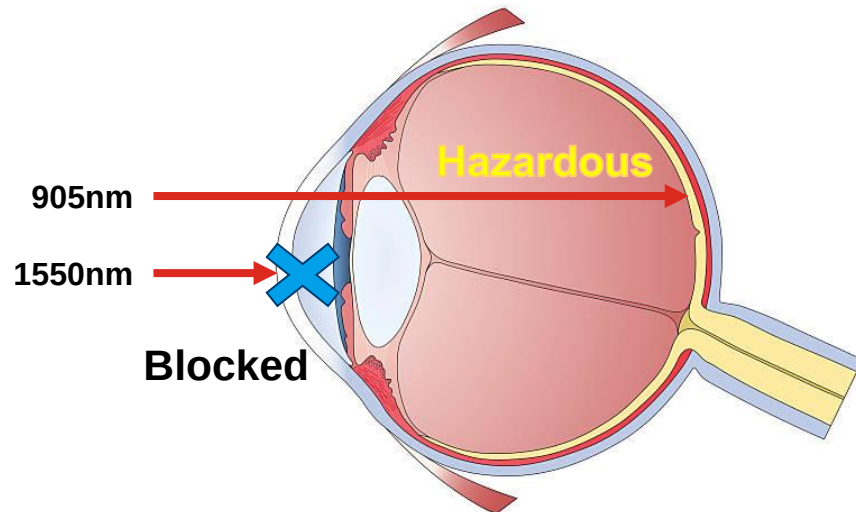
■ Comparison with RADAR

- ▶ Wavelength of LiDAR and 77GHz mm Wave
 - LiDAR: *905nm, 1550nm*
 - 77GHz: *4mm*
- ▶ Radar has **4,000 times more beam divergence** than LiDAR
- ▶ Sensors using electromagnetic waves make it as **difficult to obtain images with high resolution** as LiDAR.



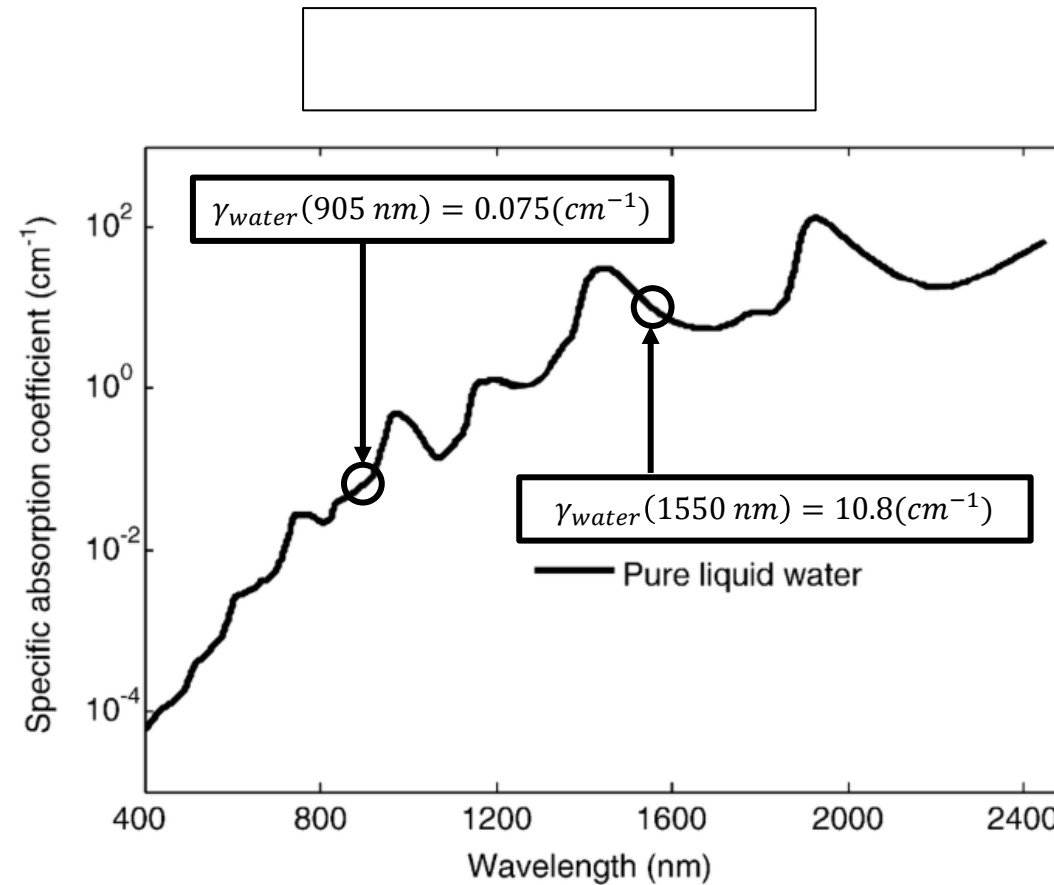
MPE (Maximum Permissible Exposure)

- High-power laser pulse can increase detection distance.
- Laser pulse should satisfy Eye safety (Maximum Permissible Exposure: MPE)!
- 1550nm laser source can be 1,000,000 times safer than 905nm.



Water Extinction Effect

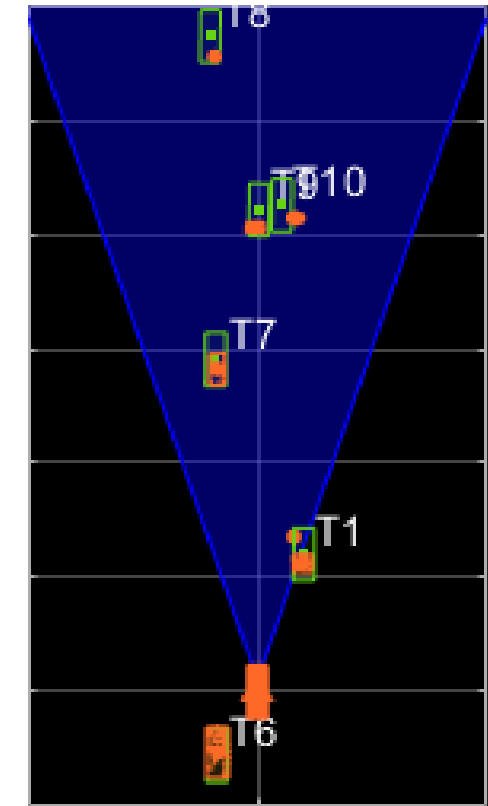
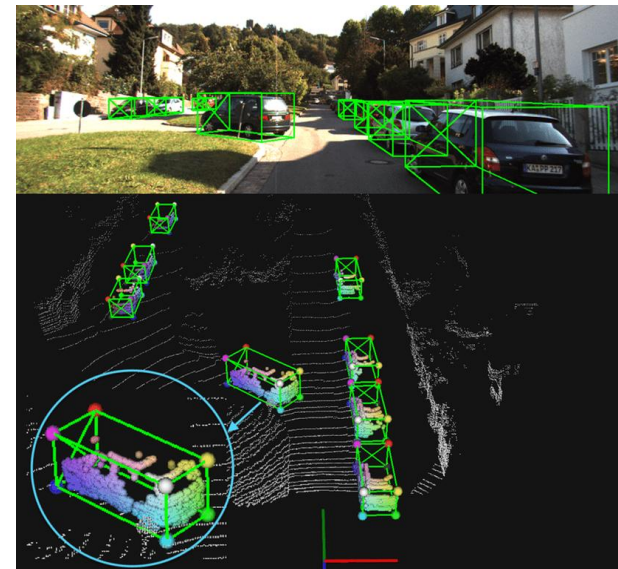
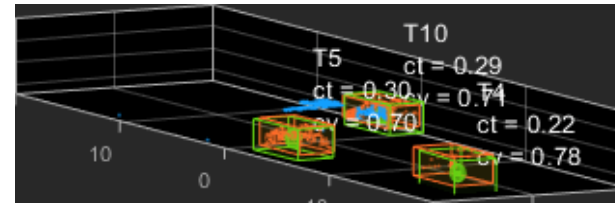
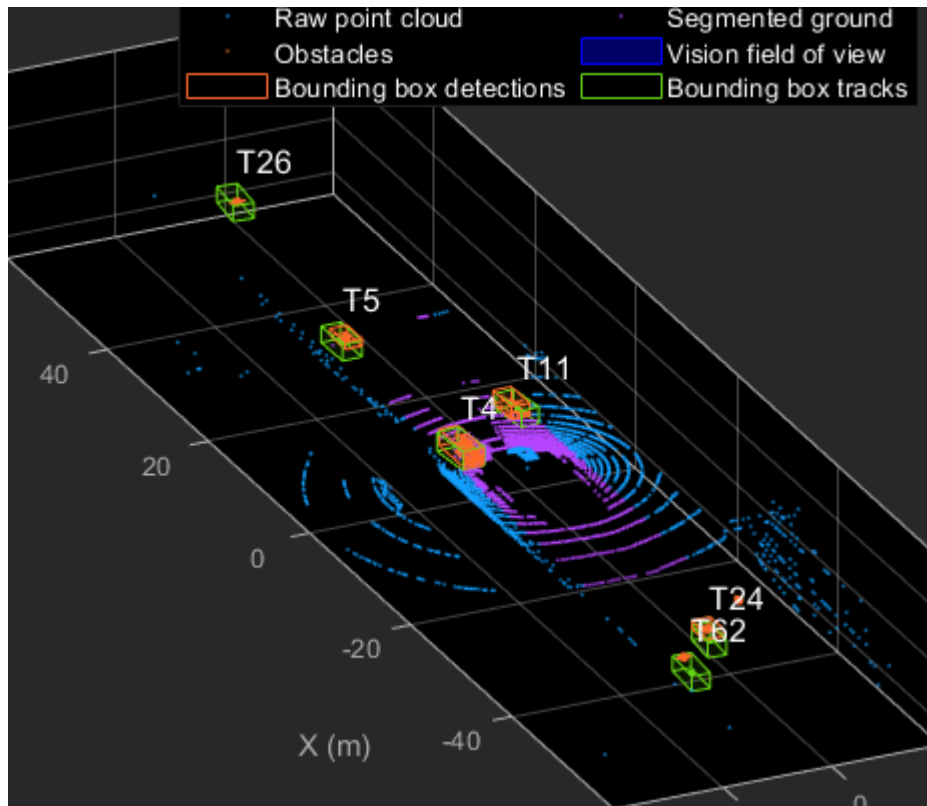
- Phenomenon where **water absorbs** and **scatters** light, **reducing the intensity** of light.
 - ▶ It occurs in the case of rain or on wet road surfaces.
- **1550nm has 140 times more extinction then 905nm.**



LiDAR based perception system

3D Object Detection

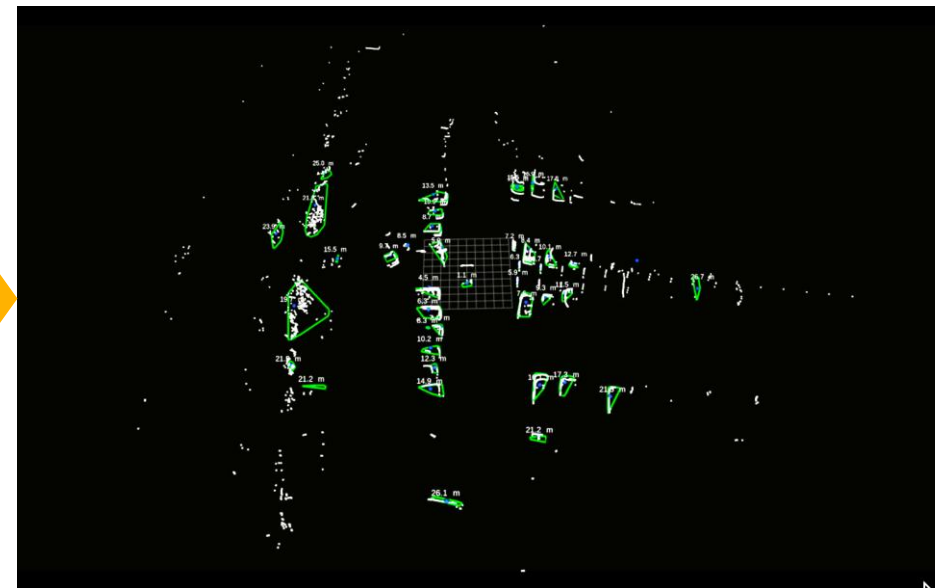
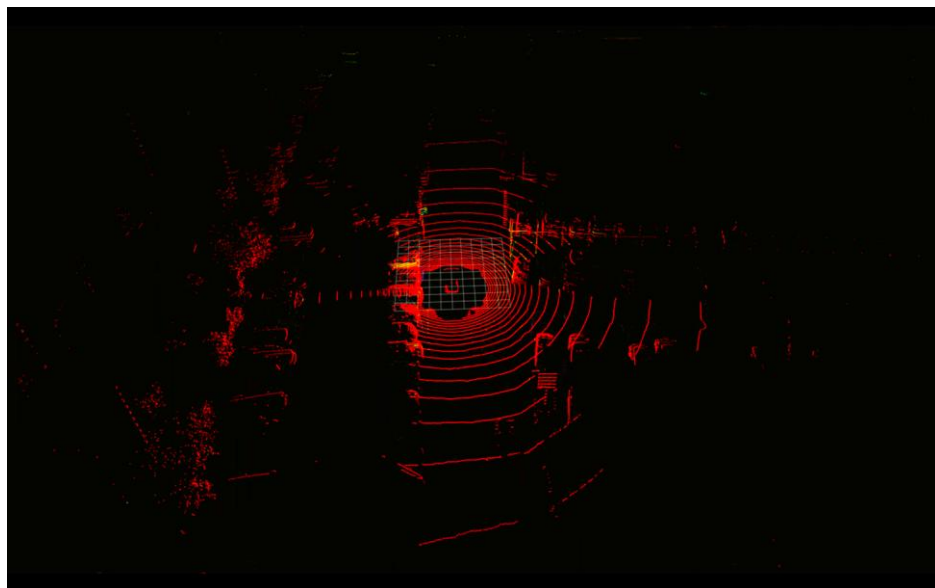
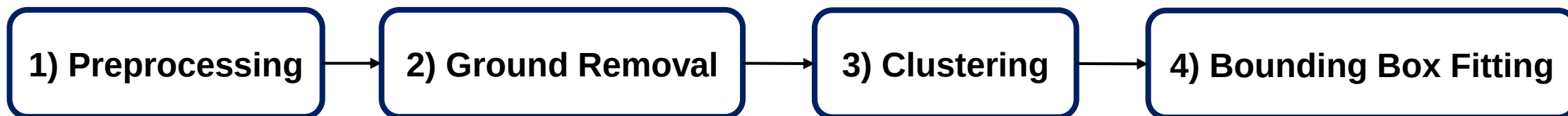
- Recognizing the position, orientation, size information of an object for a class of interest for a point cloud.
- Two approaches to detecting 3D objects using LiDAR
 - ▶ **Traditional (rule-based)** methods and **machine learning** methods.



3D Object Detection - Traditional Method

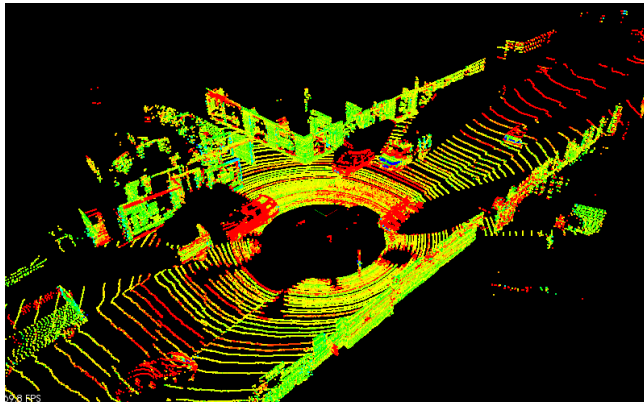
■ Pipeline

- ▶ **Preprocessing**: processing raw point cloud data to remove noise and extract useful information.
- ▶ **Ground Removal**: eliminate unnecessary points on the ground.
- ▶ **Clustering**: cluster points and detect objects.
- ▶ **Bounding Box Fitting**: fitting bounding boxes around detected objects.

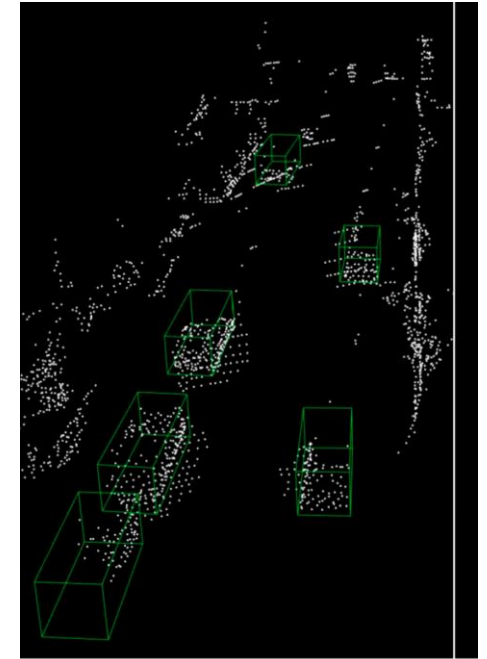
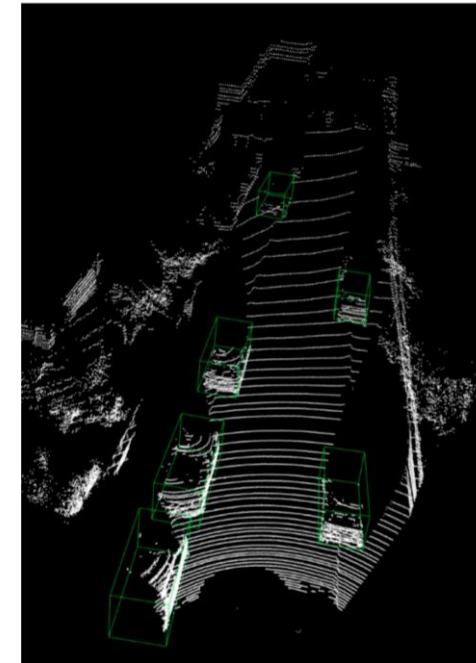
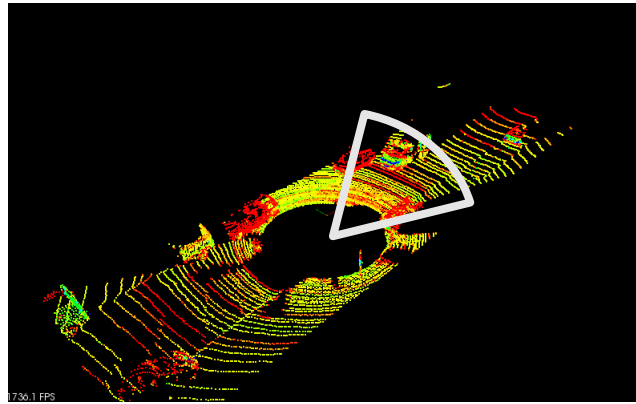


Traditional Method – Preprocessing

- Processing raw point cloud data collected from LiDAR to remove noise and extract useful information.
- Performing to make the data more suitable for subsequent analysis or application.
 - ▶ Setting ROI(Region Of Interest), Downsampling to reduce the amount of point cloud data computation is also a process of preprocessing



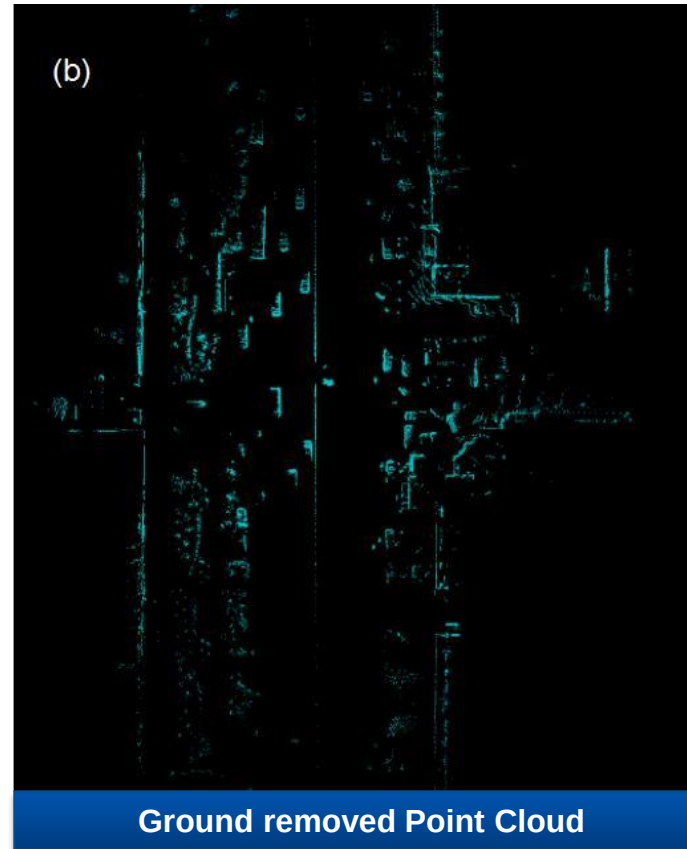
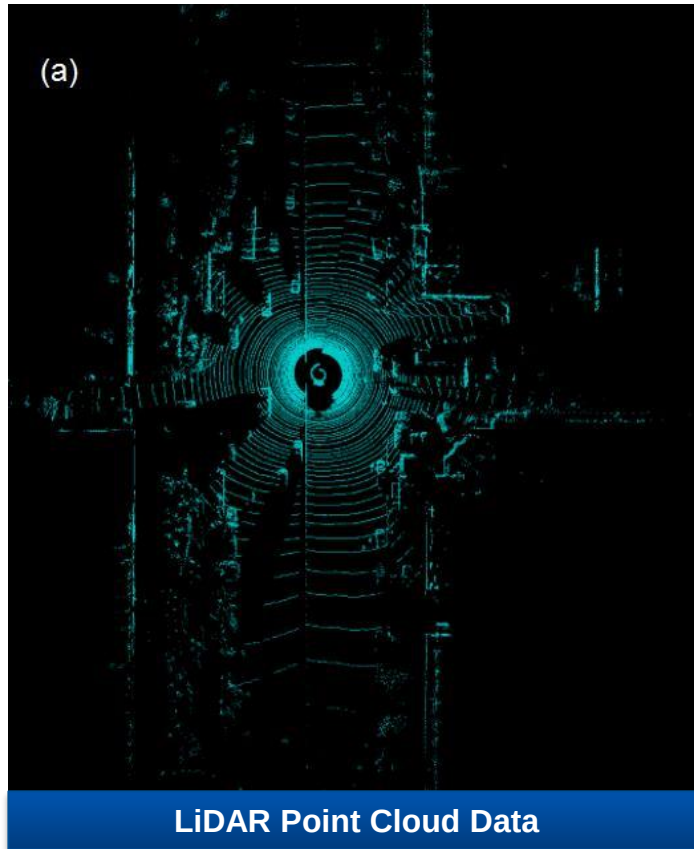
Setting ROI (Region Of Interest)



DownSampling

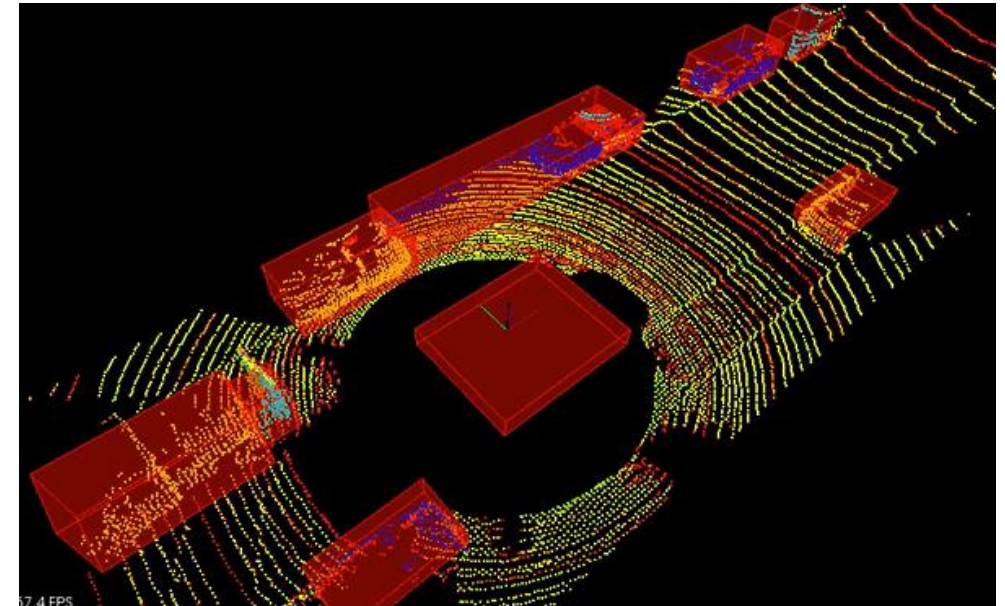
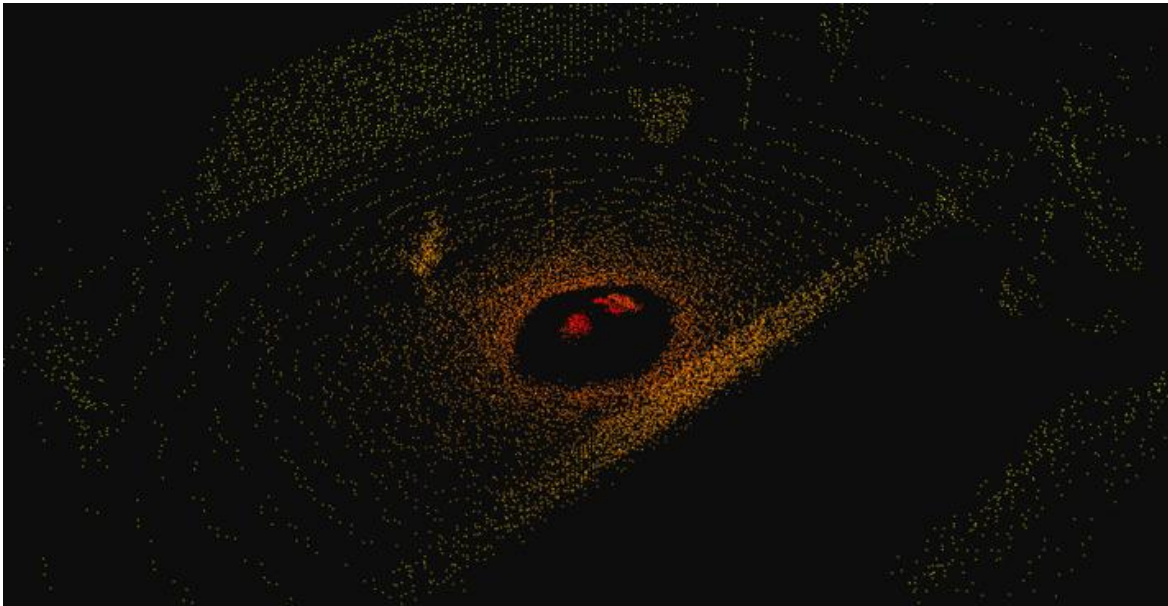
Traditional Approach – Ground Extraction

- Identifying and removing ground points from the collected point cloud.
- It is needed to detect and analyze non-ground objects such as vehicles, pedestrians, and buildings within a 3D environment.
 - ▶ RANSAC(Random Sample Consensus), Height-based thresholding etc..



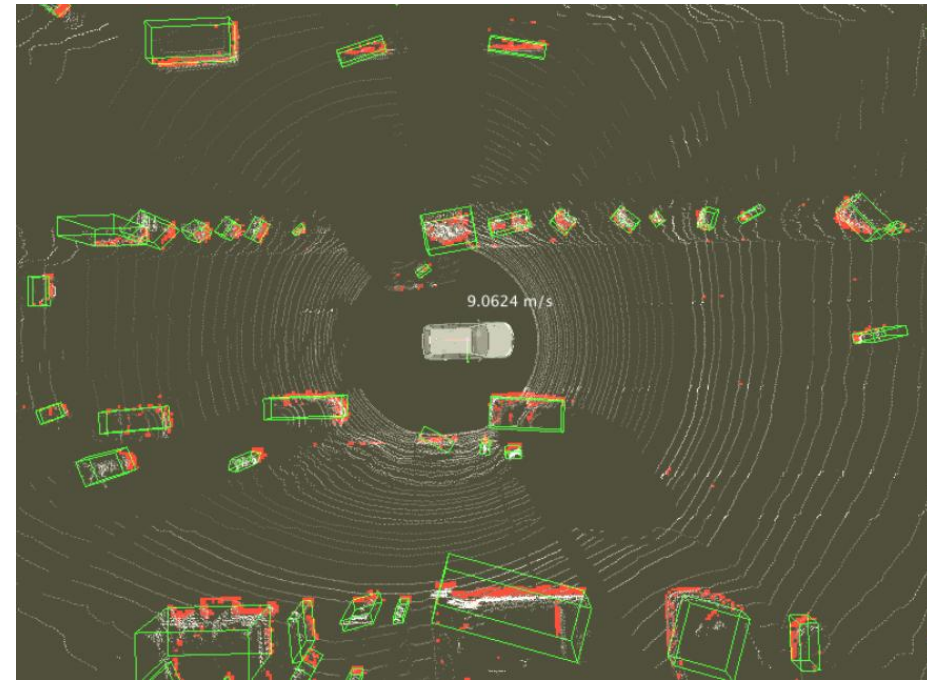
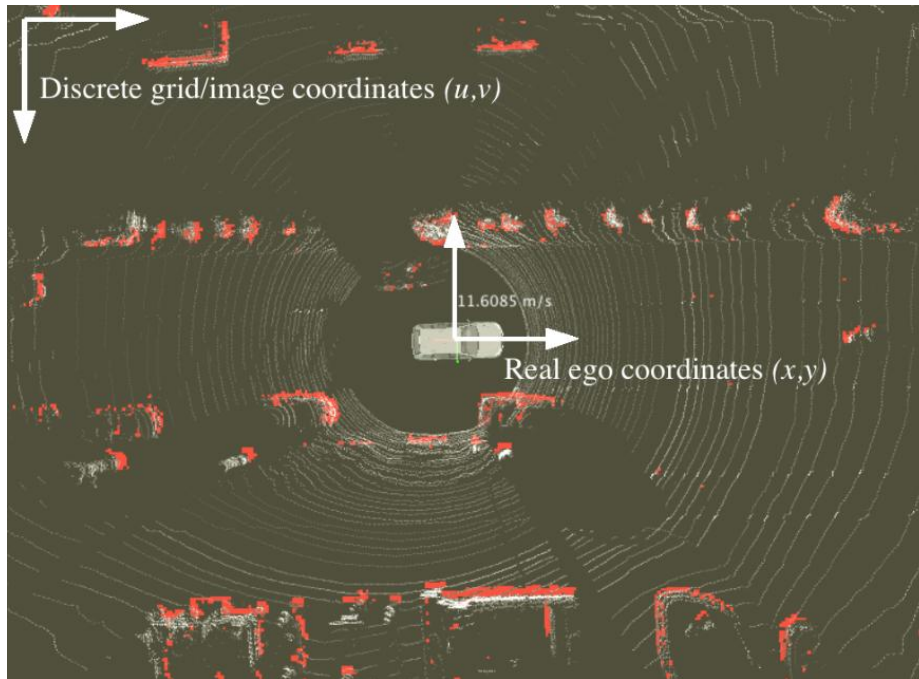
3D Object Detection – Clustering

- The definition of clustering refers to the process of dividing objects into several subgroups given objects.
- Among the traditional clustering methods, the most widely used method is clustering based on the Euclidean distance.
- Classifies points with close Euclidean Distance into one point cluster, and detects objects based on the cluster.



Traditional Method – Bounding Box Fitting

- Wrapping bounding boxes on clustered objects.
- Using to indicate the location, size, and orientation of an object and provides structural **information** needed to more easily identify and track the object.

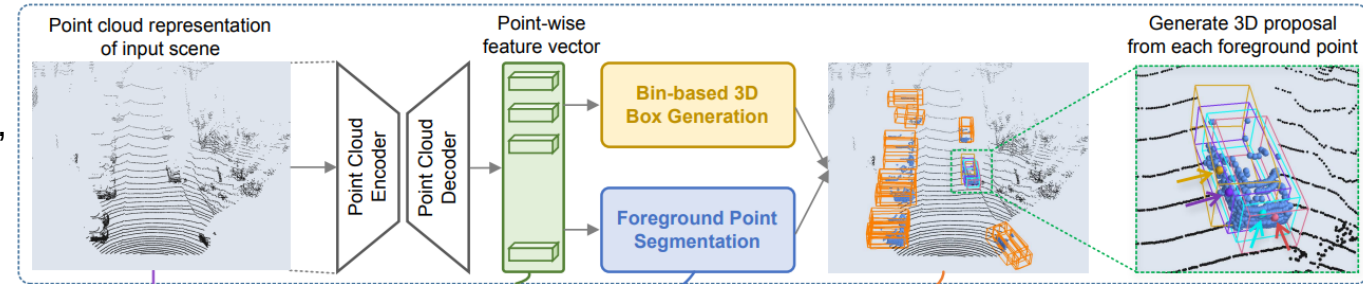


3D Object Detection – Machine Learning Method

■ 3D objects can be predicted with several machine learning techniques

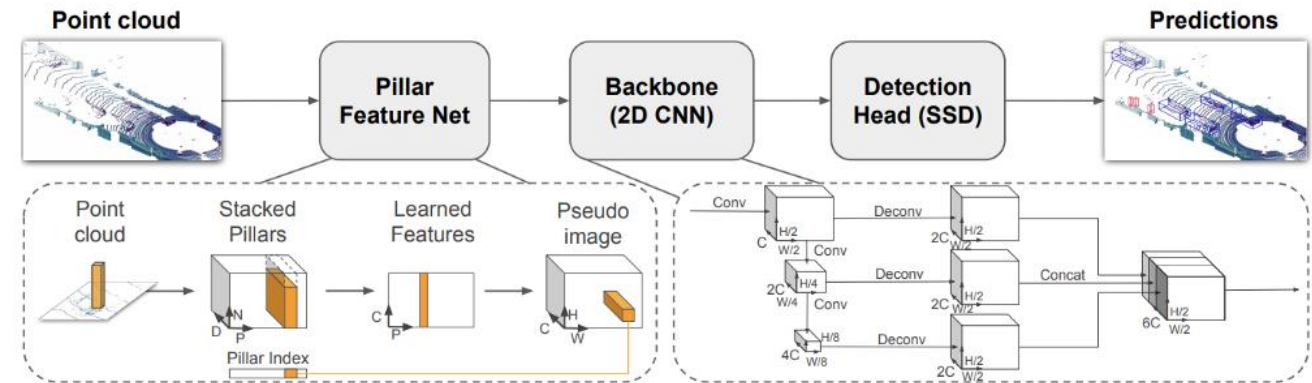
► Point-based

- Point clouds are fed directly into the network, **reducing loss of information**



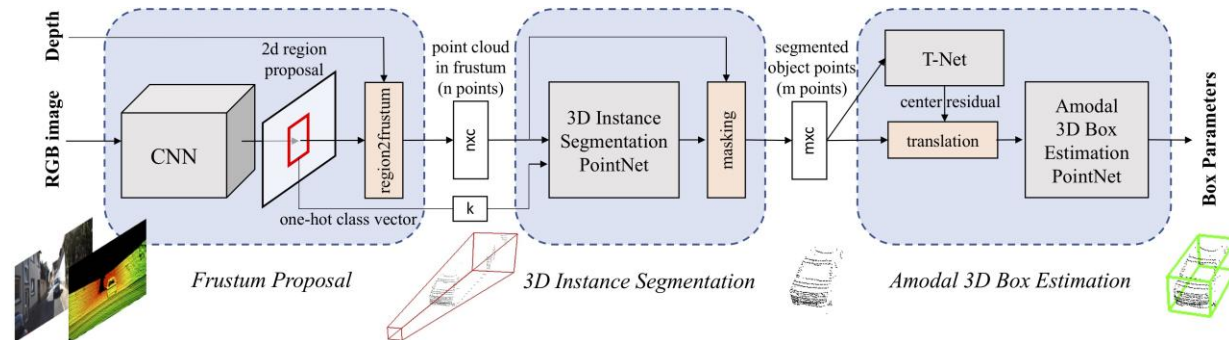
► Voxel-based

- Voxels(**Volume** + **Pixel**) are used to **reduce the amount of information** in the point cloud



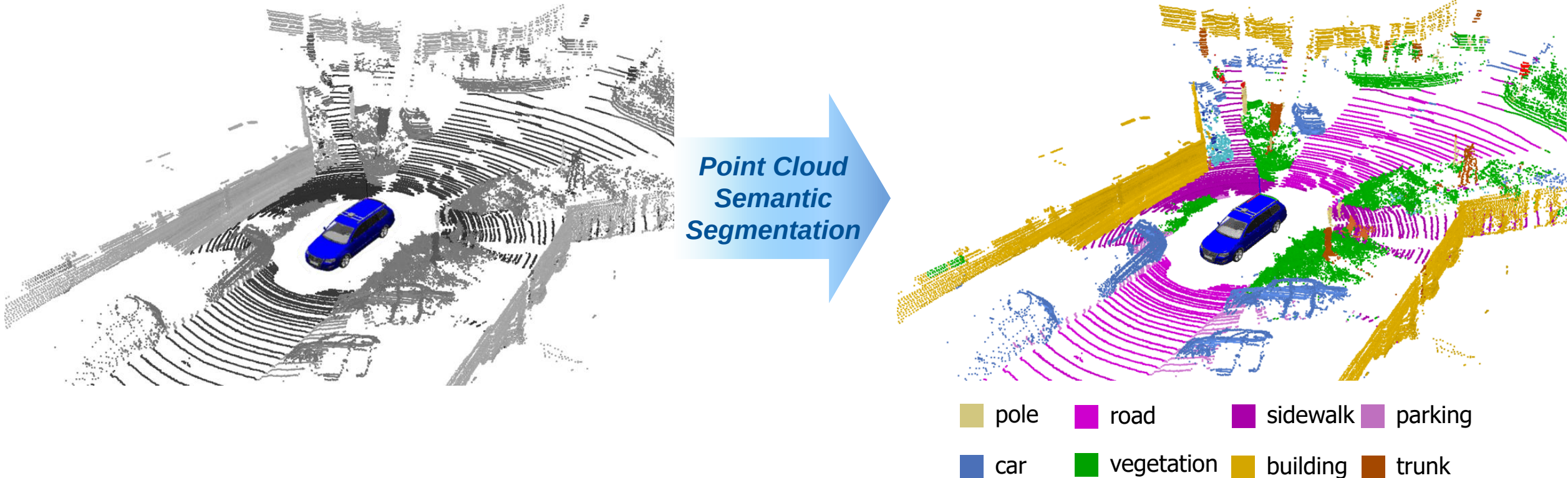
► Projection-based

- Represent point clouds data as **2D views** from different directions



Point cloud semantic segmentation

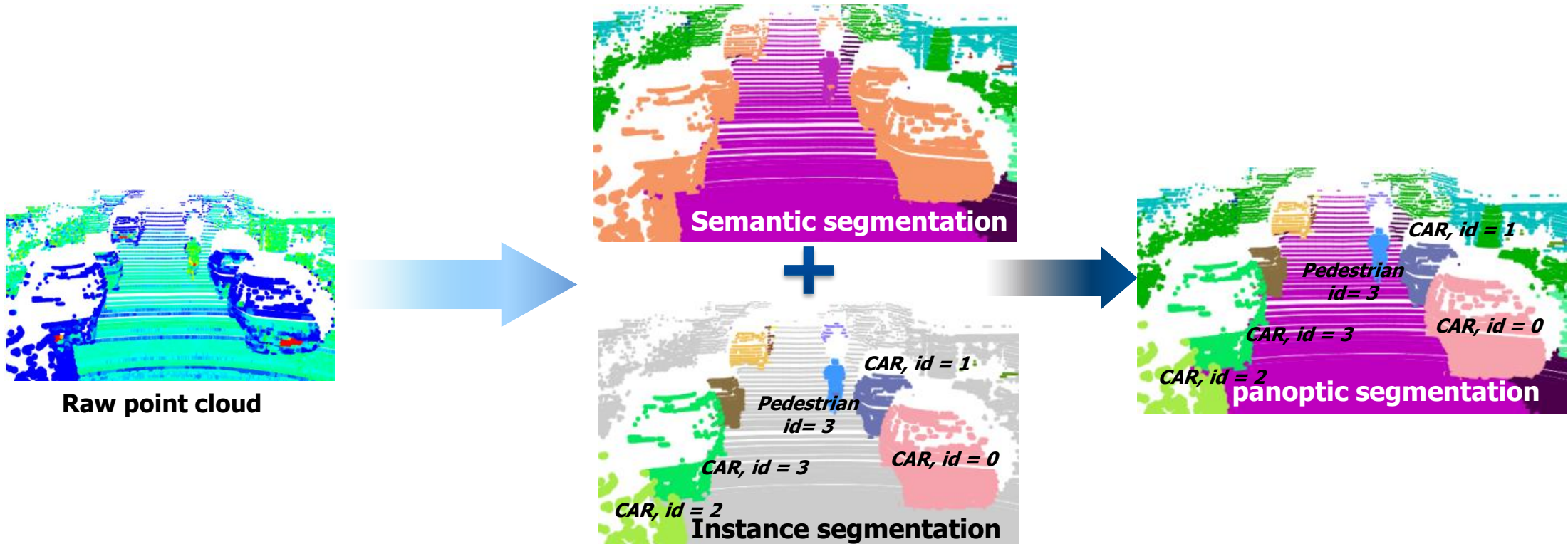
- **Point Cloud Semantic Segmentation** aims to assign class labels for each point.
 - ▶ Since it is **point-wise classification**, It helps better understand the surrounding environment.
 - ▶ Both dynamic and static objects can be classified. (cars, roads, vegetation, etc.)



Panoptic segmentation

■ Panoptic segmentation

- ▶ **Semantic** segmentation + **Instance** segmentation
 - Semantic segmentation: Predict **every point** of input into each **class** (No individual id for objects)
 - Instance segmentation: Predict on points for the **class of interest** (With individual id for objects)
- ▶ Panoptic segmentation: **Classification of all points** and **individual IDs** for interest classes



LiDAR SLAM

■ SLAM (Simultaneous Localization And Mapping)

- ▶ Technological mapping method that allows robots and other autonomous vehicles to **build a map and localize** itself on that map at the **same time**.



LIO-SAM implemented by AILAB



Velodyne 16ch LiDAR with barebone computer



**THANK YOU
FOR YOUR ATTENTION**